

Mach–Zehnder Interferometer Course Project

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Abstract—This report presents the design, fabrication, and experimental characterization of silicon-photonic Mach–Zehnder interferometers (MZIs) on a silicon-on-insulator (SOI) platform. Compact models of the effective refractive index were developed from eigenmode simulations of 500 nm single-mode and 3.0 μm multimode silicon waveguides, enabling analytical prediction of the group index, chromatic dispersion, and free spectral range (FSR). The extracted group indices and FSRs showed excellent agreement with analytical predictions and fabrication-corner simulations, confirming both the validity of the compact-model approach and the fabrication robustness. Nominally identical devices exhibited only a small effective-index variation of approximately 0.006. A hybrid MZI incorporating a 3.0 μm -wide section produced a measured group index consistent with a length-weighted prediction based on its composite waveguide geometry. These results demonstrate accurate modeling and reproducible fabrication of SOI-based interferometric devices.

I. INTRODUCTION

Silicon photonics has emerged as a key enabling technology for next-generation optical communication systems, driven by its compatibility with complementary metal–oxide–semiconductor (CMOS) fabrication processes, high integration density, and low power consumption [1], [2]. In particular, photonic integrated circuits (PICs) based on silicon-on-insulator (SOI) platforms have been widely adopted for high-speed optical interconnects in data-center and telecommunication applications [3], [4].

Among various photonic building blocks, the Mach–Zehnder interferometer (MZI) plays a central role in silicon photonics. MZIs serve as fundamental components for a wide range of functionalities, including optical modulation, switching, filtering, and phase-sensitive signal processing [5], [6]. Their broadband operation, high extinction ratio, and design flexibility make MZIs indispensable in practical transceiver architectures for optical communication systems.

Beyond conventional communication applications, MZIs have recently attracted significant attention in emerging computing paradigms such as optical accelerators and neuromorphic computing. In photonic neural networks and optical matrix–vector multiplication engines, MZIs are widely used as programmable linear transformation units that implement weighted interference and phase control [7], [8], [9]. The scalability and reconfigurability of MZI meshes have positioned them as core architectural elements for optical computing and artificial intelligence hardware accelerators.

As the scale and functionality of MZI-based circuits continue to increase, challenges related to device footprint, loss, phase errors, and fabrication tolerance become increasingly critical. In particular, compact and low-loss MZI designs are

essential to enable large-scale integration while maintaining energy efficiency and signal fidelity. These requirements motivate the exploration of new MZI design methodologies optimized for both optical performance and physical compactness.

II. THEORY OF THE MACH–ZEHNDER INTERFEROMETER

The ideal behavior of a MZI can be analytically described by deriving its transfer function. In this section, we formulate the transfer function of an ideal MZI under commonly adopted assumptions. The splitter and combiner stages of an MZI are typically implemented using either a 1×2 multimode interference (MMI) coupler, a Y-branch, or a directional coupler (DC). Figure 1 illustrates a representative MZI configuration employing Y-branches as both the splitter and the combiner. In general, phase modulation in an MZI is introduced by creating an imbalance between the two arms. This can be achieved either by introducing a path length difference ΔL between the two arms or by modulating the propagation constant β via an effective refractive index change. Such configurations are commonly referred to as imbalanced MZIs.

We assume that the input Y-branch ideally splits the optical power equally into the two arms. If the incident optical intensity and field amplitude are denoted by I_i and E_i , respectively, the optical intensities in the two arms are given by

$$I_1 = I_2 = \frac{I_i}{2}. \quad (1)$$

Correspondingly, the electric field amplitudes in each arm are

$$E_1 = E_2 = \frac{E_i}{\sqrt{2}}. \quad (2)$$

Let the lengths of the two arms be L_1 and L_2 , respectively. The corresponding propagation constants are denoted by

$$\beta_k = \frac{2\pi n_{\text{eff},k}}{\lambda} \quad (k = 1, 2), \quad (3)$$

where $n_{\text{eff},k}$ is the effective refractive index of the k -th arm and λ is the vacuum wavelength. We also introduce the power attenuation coefficients α_1 and α_2 for the two arms. The complex phase accumulation during propagation can be written as

$$\exp \left[-i \left(\beta_k - i \frac{\alpha_k}{2} \right) L_k \right]. \quad (4)$$

Therefore, the electric field at the end of each arm is expressed as

$$E_{o,k} = \frac{E_i}{\sqrt{2}} \exp \left[-i \left(\beta_k - i \frac{\alpha_k}{2} \right) L_k \right], \quad k = 1, 2. \quad (5)$$

When the two optical fields recombine at the output Y-branch

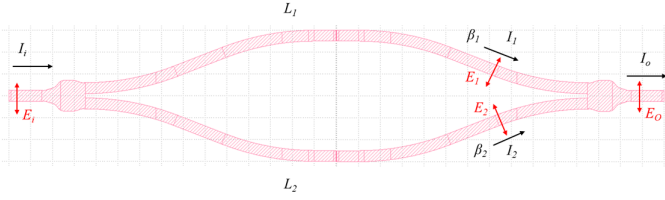


Fig. 1. Schematic of a Mach-Zehnder interferometer employing Y-branches as the splitter and combiner. An optical path length difference ΔL between the two arms, L_1 and L_2 induces wavelength-dependent interference at the output.

(or combiner), each field is again split between the fundamental mode and higher-order or radiation modes. Assuming ideal recombination into the fundamental mode, an additional factor of $1/\sqrt{2}$ applies to each arm. The total output electric field is thus given by the sum of $E_{o,1}$ and $E_{o,2}$ as

$$E_o = \frac{E_i}{2} \left[\exp \left(-i \left(\beta_1 - i \frac{\alpha_1}{2} \right) L_1 \right) + \exp \left(-i \left(\beta_2 - i \frac{\alpha_2}{2} \right) L_2 \right) \right]. \quad (6)$$

In the special case where both arms have identical propagation constants and absorption coefficients, i.e.,

$$\beta_1 = \beta_2 = \beta, \quad \alpha_1 = \alpha_2 = \alpha, \quad (7)$$

the output intensity becomes

$$I_o = \frac{I_i}{4} \left| \exp \left(-i \left(\beta - i \frac{\alpha}{2} \right) L_1 \right) + \exp \left(-i \left(\beta - i \frac{\alpha}{2} \right) L_2 \right) \right|^2. \quad (8)$$

After simplification, this yields

$$I_o = \frac{I_i}{2} \left[\frac{e^{-\alpha L_1} + e^{-\alpha L_2}}{2} + e^{-\alpha(L_1+L_2)/2} \cos(\beta \Delta L) \right], \quad (9)$$

where the path length difference is defined as

$$\Delta L = |L_1 - L_2|. \quad (10)$$

In the limit of negligible absorption ($\alpha \approx 0$), the output intensity simplifies to

$$I_o = \frac{I_i}{2} [1 + \cos(\beta \Delta L)]. \quad (11)$$

The resulting periodic modulation corresponds to the free spectral range (FSR). Expressed in terms of the propagation constant β , the FSR is given by

$$\text{FSR}(\beta) = \frac{2\pi}{\Delta L}. \quad (12)$$

Using the relation $\beta = 2\pi n_{\text{eff}}/\lambda$ and introducing the group index

$$n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}, \quad (13)$$

the FSR in the wavelength domain is obtained as

$$\text{FSR}(\lambda) = \frac{\lambda^2}{n_g \Delta L}. \quad (14)$$

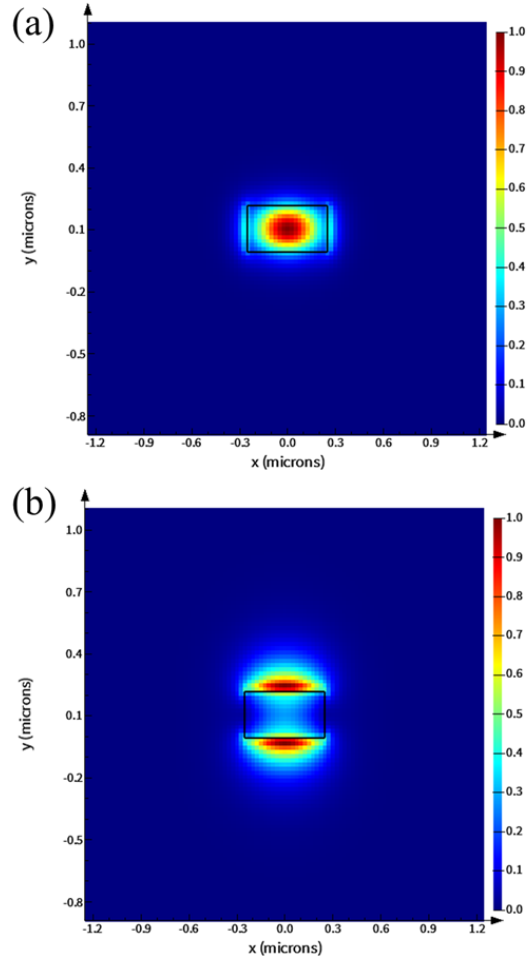


Fig. 2. Simulated optical field distributions of the fundamental (a) TE-like and (b) TM-like modes for the 500 nm-wide waveguide.

III. MODELING AND SIMULATION

A. Waveguide Design

In this section, we describe the design overview of the Si waveguides used in this study. Strip-type silicon waveguides were employed, and their dimensions were chosen to ensure single-mode operation. Specifically, the waveguide width and thickness were set to 500 nm and 220 nm, respectively. The waveguide eigenmodes and their optical properties were calculated using the MODE eigenmode solver in the Lumerical simulation suite. Figure 2 shows the simulated optical field distributions of the fundamental transverse-electric-like (TE-like) and transverse-magnetic-like (TM-like) modes for the designed waveguide.

As the waveguide width increases, higher-order modes such as the TE_2 , TM_2 are supported in addition to the fundamental modes. When the waveguide width approaches approximately 900 nm, a distinct TE_2 mode with two well-separated lobes is clearly observed. Even at a width of 600 nm, although the mode profile does not exhibit a typical TE_2 -like pattern, modes with an effective refractive index exceeding that of the SiO_2

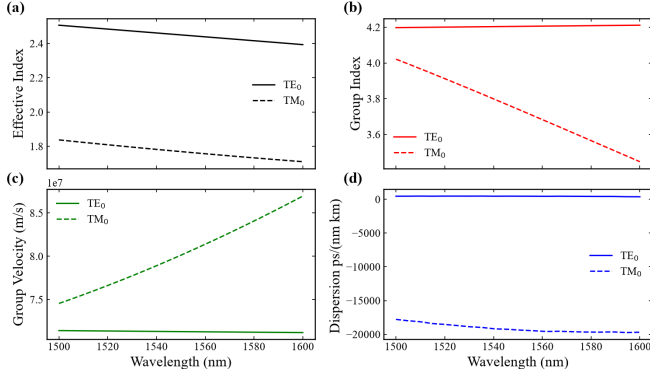


Fig. 3. Wavelength dependence of key optical properties for the single-mode waveguide with a width of 500 nm.

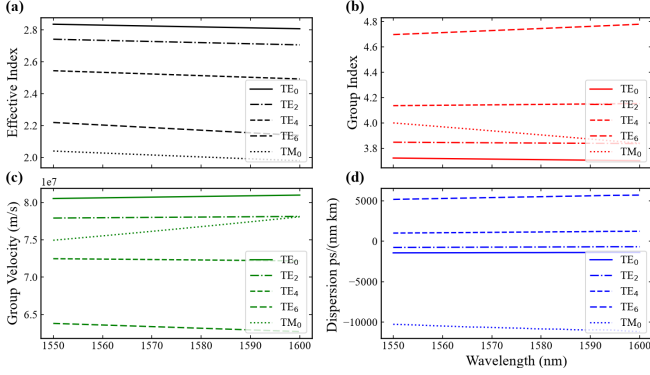


Fig. 4. Wavelength dependence of key optical properties for the multimode waveguide with a width of 3.0 μm .

cladding ($n \approx 1.44$) are present.

This indicates a potential transition to multimode operation. Therefore, a waveguide width of 500 nm was selected as the baseline design to ensure robust single-mode operation. To mitigate scattering losses arising from sidewall roughness, an additional multimode waveguide design with an increased width of 3.0 μm was also considered. For this width, higher-order modes such as TE_4 and TE_6 , characterized by four-lobed and six-lobed field distributions, respectively, are supported.

Figure 3 and Figure 4 summarize the wavelength dependence of the effective refractive index, group index, group velocity, and dispersion for the 500 nm-wide and 3.0 μm -wide waveguides, respectively. The chromatic dispersion is expressed using the group-velocity dispersion (GVD) parameter β_2 as

$$D = -\frac{2\pi c}{\lambda^2} \beta_2, \quad (15)$$

where c is the speed of light in vacuum. The GVD is related to the group velocity v_g as

$$\beta_2 = \frac{d}{d\omega} \left(\frac{1}{v_g} \right), \quad (16)$$

with the group velocity given by

$$v_g = \frac{c}{n_g}, \quad (17)$$

TABLE I
FITTING PARAMETERS OF THE COMPACT MODEL FOR THE 500 nm WIDE SINGLE-MODE WAVEGUIDE.

Parameter	TE_0	TM_0
n_0	2.4502	1.7692
n_1	-1.1320	-1.2721
n_2	-0.04416	1.8616

TABLE II
FITTING PARAMETERS OF THE COMPACT MODEL FOR THE 3.0 μm -WIDE MULTIMODE WAVEGUIDE.

Parameter	TE_0	TM_0
n_0	2.8029	1.9953
n_1	-0.6237	-1.2895
n_2	0.1183	1.0223

where n_g denotes the group refractive index. Consequently, the dispersion is directly proportional to the wavelength derivative of the group refractive index.

A compact models for the effective refractive index for the single-mode and multimode waveguides were constructed by performing a least-squares fit to the simulated effective index data shown in Fig. 3 and Fig. 4, respectively. The effective refractive index was approximated by a second-order Taylor expansion around a reference wavelength $\lambda_0 = 1550$ nm as

$$n_{\text{eff}}(\lambda) = n_0 + n_1(\lambda - \lambda_0) + n_2(\lambda - \lambda_0)^2. \quad (18)$$

The coefficient of determination (R^2) was found to be approximately unity, indicating excellent agreement between the compact model and the numerical simulation. The extracted fitting parameters for the single-mode and multimode waveguides are summarized in Table I and Table II, respectively.

B. Bending Radius Design

The bending radius was investigated for both single-mode and multimode waveguides. The bending loss was evaluated by calculating the squared overlap integral of the electric field distributions between a straight waveguide and a bent waveguide, which represents the loss induced by mode mismatch. Although radiation loss due to outward leakage in the radial direction is also present, the dominant contribution to the bending loss originates from mode mismatch in the present waveguide configurations.

In addition to the bending-induced loss, material absorption and scattering losses were assumed to be constant and independent of the bending radius. These contributions were set to 2 dB/cm for the single-mode waveguide and 1 dB/cm for the multimode waveguide. The simulation results are summarized in Fig. 5.

To suppress the total bending loss to a negligible level on the order of 0.01 dB or lower, the bending radii were determined based on the results shown in Fig. 5(a) and Fig. 5(b). For the single-mode waveguide, bending radii of $r = 7$ μm and $r = 15$ μm were adopted for the TE_1 and TM_1 modes, respectively.

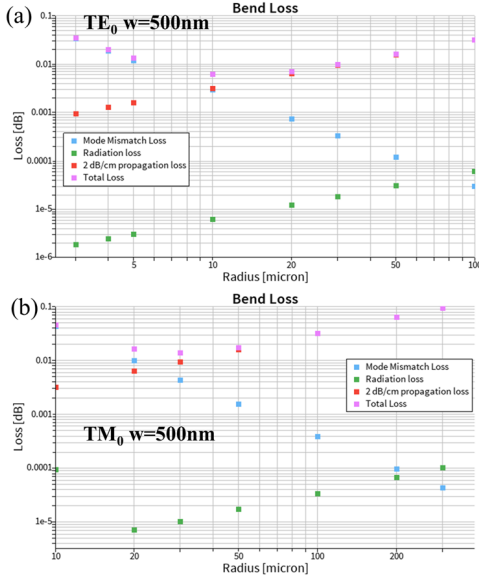


Fig. 5. Simulated bending loss as a function of bending radius. (a) TE_1 mode and (b) TM_1 mode in the single-mode waveguide. The bending loss is evaluated based on the squared overlap integral between the field distributions of straight and bent waveguides, including the assumed contributions of material absorption and scattering losses.

TABLE III
CALCULATED GROUP INDEX n_g AT $\lambda_0 = 1550$ NM DERIVED FROM THE COMPACT MODEL FITTING COEFFICIENTS.

Waveguide	TE_0	TM_0
500 nm (single-mode)	4.20	3.74
3.0 μm (multimode)	3.77	3.99

TABLE IV
CALCULATED FSR AT $\lambda_0 = 1550$ NM FOR DIFFERENT PATH LENGTH DIFFERENCES ΔL .

ΔL (μm)	FSR (nm)	
	TE_0 (500 nm)	TM_0 (500 nm)
25	22.9	25.7
50	11.4	12.9
100	5.71	6.43
200	2.86	3.21
400	1.42	1.61

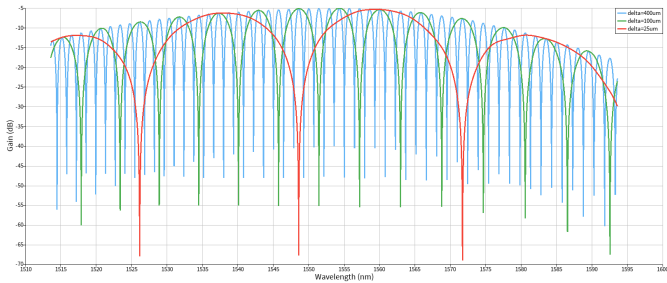


Fig. 6. Simulated transmission spectra of MZIs designed using the EBeam PDK. The path length differences were $\Delta L = 25, 100$, and $400 \mu\text{m}$.

TABLE V
SUMMARY OF THE TEG DEVICES USED FOR THE EXPERIMENTAL EVALUATION.

Label	Device	Pol.	Band	Width (μm)	ΔL (μm)
Ref	WG loopback	TE	C	0.5	—
MZI1	MZI	TE	C	0.5	25
MZI2	MZI	TE	C	0.5	25
MZI3	MZI	TE	C	0.5	100
MZI4	MZI	TE	C	0.5	200
MZI5	MZI	TE	C	0.5	400
MZI6	MZI	TM	C	0.5	25
MZI7	MZI	TM	C	0.5	100
MZI8	MZI	TE	C	3.0/0.5	400
ADC	Adiabatic DC	TE	C	0.5	470
BDC	Broadband DC	TE	C	0.5	470

C. MZI Design

From the compact model, the group index n_g at the reference wavelength $\lambda_0 = 1550$ nm can be analytically derived from the fitting coefficients, given by

$$n_g(\lambda_0) = n_0 - \lambda_0 n_1. \quad (19)$$

The group indices for the single-mode and multimode waveguides were calculated from the fitted coefficients listed in Tables I and II. The resulting group indices at $\lambda_0 = 1550$ nm are summarized in Table III.

The FSR of the MZI was calculated using Eq. (14). Table IV summarizes the calculated FSR values for different path length differences ΔL for both single-mode and multimode waveguides.

MZI simulations were carried out using the EBeam process design kit (PDK) provided by the University of British Columbia (UBC). The simulations were performed with the INTERCONNECT module of the Lumerical simulation suite. Y-branches were employed as both the splitter and the combiner, and simulations were conducted for single-mode waveguides with path length differences of $\Delta L = 25, 100$, and $400 \mu\text{m}$. All simulations focused on the TE polarized mode.

The simulated transmission spectra for the TE mode are shown in Fig. 6. For example, the FSRs around the central wavelength $\lambda_0 = 1550$ nm were found to be 1.4 nm, 5.6 nm, and 21.6 nm for $\Delta L = 25, 100$, and $400 \mu\text{m}$, respectively. These values are in good agreement with the analytical predictions based on the expressions summarized in Table IV.

IV. DESIGN OF TEST STRUCTURES

To experimentally characterize the FSR and the group index n_g behavior of the MZIs and directional couplers (DCs) discussed above, a dedicated test element group (TEG) was designed and included on the fabricated chip. The evaluation was carried out for the TE polarization over the C-band unless otherwise noted, using the device set summarized in Table V. Figure 7 shows the layout of the fabricated TEG. The structures were arranged to allow direct comparison between devices having different path-length differences, waveguide widths, and coupler configurations.

A plain waveguide loopback (Ref) was included as a reference structure to calibrate the fiber-to-chip coupling loss and

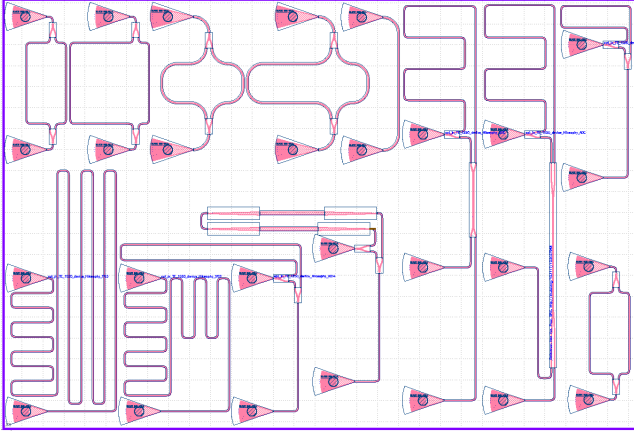


Fig. 7. Layout of the fabricated test element group (TEG). The chip includes reference loopback waveguides, MZIs with various path-length differences, adiabatic and broadband directional couplers, and the hybrid-waveguide MZI8. In MZI8, the differential path consists of both a widened $3.0\ \mu\text{m}$ multimode waveguide section and conventional $0.5\ \mu\text{m}$ single-mode waveguides connected through adiabatic tapers.

to provide a baseline transmission spectrum for normalization. MZI1 and MZI2 share an identical layout with $\Delta L = 25\ \mu\text{m}$ and were placed at different locations on the same chip so that intra-chip variations in the measured FSR and extinction ratio could be assessed. MZI3 through MZI5 retain the same waveguide cross-section but employ progressively larger path length differences ($\Delta L = 100, 200,$ and $400\ \mu\text{m}$, respectively), allowing the wavelength dependence of n_g to be extracted from the measured FSR at each ΔL and compared with the compact-model prediction of Eq. (14). MZI6 was added as a further $\Delta L = 25\ \mu\text{m}$ device to supplement the statistical comparison with MZI1 and MZI2.

To investigate the impact of the waveguide cross-section on the group-index behavior and to mitigate propagation loss caused by sidewall roughness, MZI8 was designed with a locally widened $3.0\ \mu\text{m}$ multimode section inserted into one arm of an otherwise standard $0.5\ \mu\text{m}$ single-mode MZI. The device maintains TE polarization and a total path-length difference of $\Delta L = 400\ \mu\text{m}$. Within the differential path, the widened multimode section has a physical length of $117.8\ \mu\text{m}$ and is connected through two adiabatic tapers having a combined length of $200\ \mu\text{m}$. The remaining section of the differential path retains the standard $0.5\ \mu\text{m}$ -wide single-mode waveguide. Consequently, the measured interference spectrum of MZI8 reflects the weighted contribution of both waveguide types rather than the properties of an isolated $3.0\ \mu\text{m}$ multimode waveguide.

In addition to the MZIs, an adiabatic directional coupler (ADC) and a broadband directional coupler (BDC), both with a coupling length corresponding to $\Delta L = 470\ \mu\text{m}$, were included to study how the coupling-ratio wavelength dependence of each DC type influences the position and depth of the interference dips in the transmission spectrum.

TM-polarized devices were implemented in MZI6 and MZI7 to extend the evaluation beyond the TE mode. However,

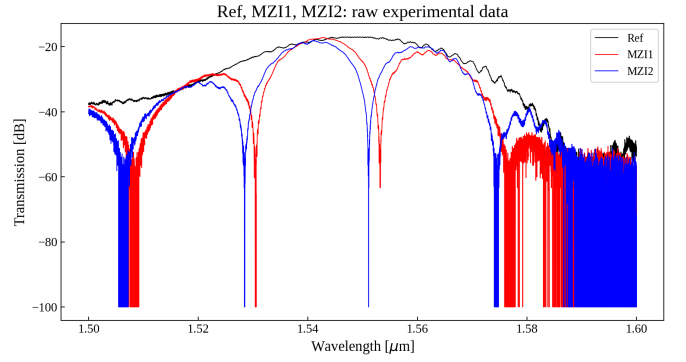


Fig. 8. Raw (uncorrected) experimental transmission spectra of the reference waveguide loopback (Ref, black) and the two nominally identical $\Delta L = 25\ \mu\text{m}$ MZIs, MZI1 (red) and MZI2 (blue), used to assess intra-chip variation.

since a grating coupler (GC) suitable for efficient TM-mode fiber coupling was not properly selected for this chip, the collected spectra for these devices did not provide sufficient signal-to-noise ratio for reliable analysis. Consequently, the experimental characterization of the TM-mode devices could not be completed in the present study.

V. DEVICE FABRICATION

The designed MZI devices were fabricated by Applied Nanotools Inc. (Edmonton, Canada) through its NanoSOI multi-project wafer (MPW) service, which is based on a direct-write $100\ \text{keV}$ electron-beam lithography (EBL) process. The starting material was a $200\ \text{mm}$ -diameter SOI wafer with a $220\ \text{nm}$ -thick device layer and a $2\ \mu\text{m}$ buried oxide, pre-diced into $25 \times 25\ \text{mm}$ chips with backside scribe lines to allow the chips to be separated cleanly after processing.

Prior to patterning, the substrates were cleaned in a piranha solution ($3:1\ \text{H}_2\text{SO}_4:\text{H}_2\text{O}_2$) for $15\ \text{min}$, rinsed in water and IPA, and then coated with hydrogen silsesquioxane (HSQ) resist, which was baked to drive off residual solvent. The patterns were written on a JEOL JBX-8100FS EBL system at the University of British Columbia.

After exposure, the resist was developed in a tetramethylammonium hydroxide (TMAH)-based developer. The patterned chips were then mounted on a 4-inch carrier wafer and etched by anisotropic ICP-RIE using a chlorine-based chemistry, with the etch rate calibrated beforehand. The remaining HSQ resist was subsequently stripped with a $10:1$ buffered oxide etch. Finally, a $2.2\ \mu\text{m}$ -thick oxide upper cladding was deposited by plasma-enhanced chemical vapor deposition (PECVD) using a tetraethyl orthosilicate (TEOS) precursor at $300\ ^\circ\text{C}$.

VI. RESULTS AND DISCUSSION

A. Transmission-Spectrum Fitting and Model Validation

As a representative example of the raw measured data, Fig. 8 shows the transmission spectra of the reference loopback waveguide (Ref) together with the two nominally identical MZIs, MZI1 and MZI2, before any baseline correction has been applied. In all of the analysis that follows, the raw spectra

TABLE VI
EXTRACTED FITTING PARAMETERS FOR THE MZI
TRANSMISSION-SPECTRUM FITS.

Device (ΔL , μm)	n_{eff} (λ_0)	n_2	n_3	α (mm^{-1})	n_g (λ_0)	FSR(λ_0) (nm)	$D(\lambda_0)$ (ps/nm/km)	R^2
MZI1 (25)	2.4577	-1.097	-0.515	4.08	4.157	23.12	-5322	0.751
MZI2 (25)	2.4520	-1.116	-0.157	4.29	4.182	22.98	-1622	0.633
MZI3 (100)	2.4499	-1.126	-0.038	1.11	4.196	5.73	397	0.789
MZI4 (200)	2.4572	-1.122	-0.036	1.00	4.196	2.86	377	0.755
MZI5 (400)	2.4514	-1.125	-0.042	0.37	4.195	1.43	432	0.796
MZI8 (400)	2.8026	-0.682	0.078	0.59	3.859	1.56	-802	0.818

TABLE VII
CORNER ANALYSIS OF THE SIMULATED GROUP INDEX AND
CORRESPONDING FSR AT $\lambda_0 = 1550$ NM

Thickness t (nm)	Width w (nm)	n_g TE	n_g TM	FSR TE (nm)	FSR TM (nm)
220.0	500	4.205	3.742	5.71	6.42
223.1	470	4.262	3.761	5.64	6.39
223.1	510	4.187	3.849	5.74	6.25
215.3	470	4.256	3.498	5.65	6.87
215.3	510	4.182	3.603	5.75	6.67

are first baseline-corrected and subsequently fitted to the ideal MZI transfer function of Eq. (11), in which the propagation constant β depends on n_{eff} and is therefore chromatically dispersive. $n_{\text{eff}}(\lambda)$ is described by the compact model of Eq. (18), so that the primary fitting parameters are the Taylor coefficients n_1 , n_2 , and n_3 together with the per-arm loss-imbalance coefficient α .

Rather than normalizing each spectrum against the Ref loopback, the baseline of each MZI spectrum was instead removed by fitting a low-order polynomial directly to the device's own spectrum; this was found to improve the accuracy of the subsequent nonlinear fit compared with Ref-based normalization. To make the fit robust against the choice of initial parameters, the free spectral range was first estimated approximately from the autocorrelation of the baseline-corrected spectrum, which in turn constrains the initial guess for n_g and, correspondingly, for n_2 . This autocorrelation-assisted initialization allows the nonlinear least-squares fit to converge reliably without requiring careful manual tuning of the initial parameters. The extracted fitting parameters for all measured MZIs are summarized in Table VI. $n_{\text{eff}}(\lambda_0) = n_1$, n_2 and n_3 are the first- and second-order Taylor coefficients of $n_{\text{eff}}(\lambda)$ about $\lambda_0 = 1.550 \mu\text{m}$, α is the fitted per-arm loss-imbalance coefficient, $n_g(\lambda_0)$ and $D(\lambda_0)$ are the group index and chromatic dispersion evaluated at $\lambda_0 = 1550$ nm, and R^2 is the coefficient of determination of the fit. Figures 9–11 show the baseline-corrected experimental data together with the corresponding fitted curves for the (MZI2, MZI3), (MZI4, MZI5), and (MZI5, MZI8) device pairs, respectively. Although the R^2 values reported in Table VI are not particularly high, the fitted curves are seen to reproduce the measured fringe pattern closely by visual inspection. Moreover, the values of n_g and FSR obtained from the fit agree closely with the values predicted analytically in Tables III and IV, confirming the validity of the extracted compact-model parameters.

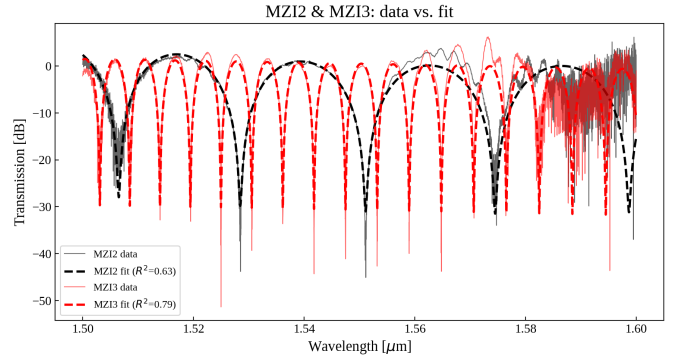


Fig. 9. Measured transmission spectra (solid) and fitted model spectra (dashed) for MZI2 ($\Delta L = 25 \mu\text{m}$) and MZI3 ($\Delta L = 100 \mu\text{m}$). The shorter fringe period of MZI3 reflects its larger path-length difference, consistent with the $\text{FSR} \propto 1/\Delta L$ relation of Eq. (14).

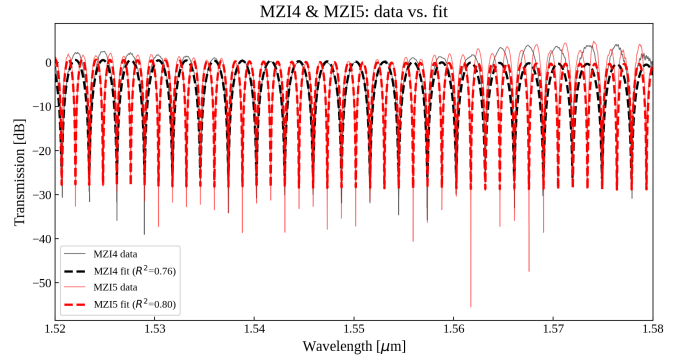


Fig. 10. Measured transmission spectra (solid) and fitted model spectra (dashed) for MZI4 ($\Delta L = 200 \mu\text{m}$) and MZI5 ($\Delta L = 400 \mu\text{m}$), shown over the 1.52–1.58 μm range for clarity.

B. Fabrication Tolerance and Intra-Chip Variation

To further evaluate the robustness of the design against fabrication-induced variations, a corner analysis was performed by varying both the waveguide thickness and width around the nominal cross-section, as summarized in Table VII. For the nominal design of $t = 220$ nm and $w = 500$ nm, the TE-mode group index and FSR for $\Delta L = 100 \mu\text{m}$ are calculated to be $n_g = 4.205$ and $\text{FSR} = 5.71$ nm, respectively (the same values as in Tables III and IV). These values are in excellent agreement with the experimentally extracted values for MZI3, namely $n_g = 4.196$ and $\text{FSR} = 5.73$ nm in Table VI. The calculated TE-mode group indices at the process corners are within a narrow range of $n_g = 4.182$ – 4.262 , while the corresponding FSR varies from 5.64 to 5.75 nm. Therefore, the measured n_g and FSR are not only close to the nominal design values but also lie well within the range predicted by the corner analysis. This agreement supports that the device patterns were generated appropriately and that the fabricated structures are consistent with the intended design within the expected process-variation window.

We next examine MZI1 and MZI2, which share nominally identical designs. As seen in Fig. 8, the transmission dips of

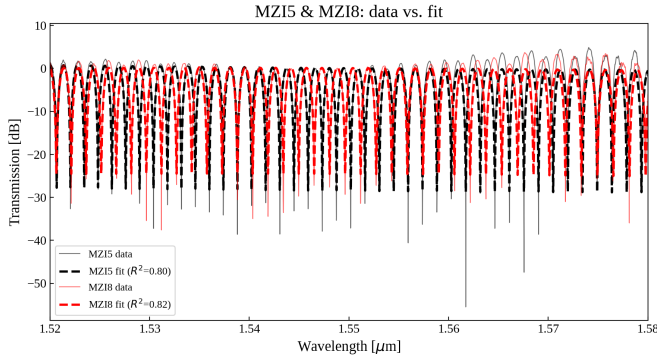


Fig. 11. Measured transmission spectra (solid) and fitted model spectra (dashed) for MZI5 ($\Delta L = 400 \mu\text{m}$, 500 nm single-mode waveguide) and MZI8 ($\Delta L = 400 \mu\text{m}$, 3.0 μm tapered multimode waveguide), shown over the 1.52–1.58 μm range for clarity.

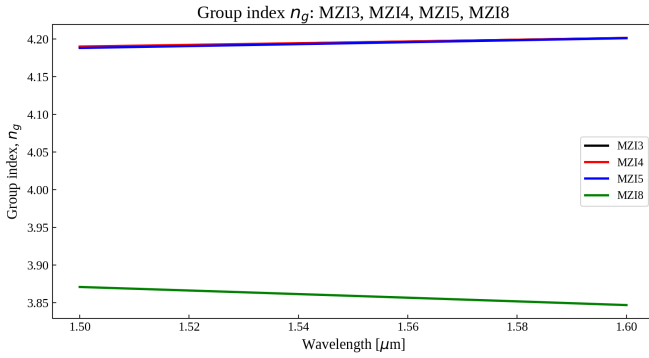


Fig. 12. Group index $n_g(\lambda)$ extracted from the MZI fits for MZI3, MZI4, MZI5 (500 nm single-mode waveguide) and MZI8 (3.0 μm tapered multimode waveguide). MZI3–MZI5 agree closely despite their different ΔL , while MZI8 shows a distinctly lower n_g consistent with the wider waveguide cross-section.

MZI1 and MZI2 near 1550 nm are offset from one another by approximately 2.1 nm. According to Table VI, this shift corresponds to a difference in n_{eff} of only about 0.006, which we attribute to intra-chip fabrication variation between the two nominally identical devices. Because these two devices have the smallest ΔL (25 μm) among all the MZIs studied, their fringe period is comparatively large, and the fit is correspondingly less well constrained; the extracted FSR values show larger scatter, with the associated variation in n_g amounting to approximately 0.025 between the two devices.

C. Group Index and Dispersion Extraction

Since the fitted compact model directly provides n_{eff} as an analytic function of λ , both the group index $n_g(\lambda)$ (Eq. (13)) and the chromatic dispersion $D(\lambda)$ (Eq. (15)) can be evaluated as continuous functions of wavelength rather than at a single point. The resulting $n_g(\lambda)$ and $D(\lambda)$ curves are shown in Fig. 12 and Fig. 13, respectively. For the single-mode 500 nm waveguide devices (MZI3, MZI4, and MZI5), the extracted $n_g(\lambda)$ curves are found to coincide almost exactly, which indicates that these nominally identical waveguides

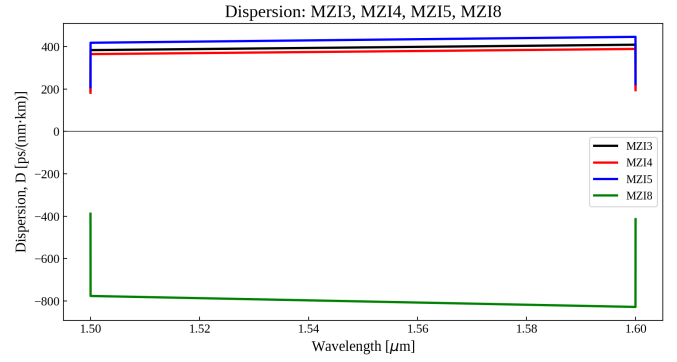


Fig. 13. Chromatic dispersion $D(\lambda)$ derived from the fitted group index for MZI3, MZI4, MZI5, and MZI8. The 3.0 μm multimode waveguide (MZI8) exhibits both a different sign and a substantially larger magnitude of dispersion compared with the 500 nm single-mode devices.

were fabricated with excellent reproducibility. The wavelength dependence of n_g also follows the same qualitative trend as that obtained from the eigenmode simulations in Fig. 3, and the extracted dispersion is of order $D \approx 400 \text{ ps}/(\text{nm}\cdot\text{km})$, in good agreement with the simulated value.

D. Hybrid Multimode MZI Analysis

MZI8 differs from the other devices in that only a portion of the differential path is widened to 3.0 μm , while the remaining sections retain the standard 0.5 μm single-mode waveguide cross-section. As described in Section IV, the $\Delta L = 400 \mu\text{m}$ path-length difference consists of a 117.8 μm -long 3.0 μm waveguide section, two adiabatic tapers with a combined length of 200 μm , and a remaining 82.2 μm section of 0.5 μm waveguide.

The compact model predicts group indices of approximately $n_g = 4.20$ and $n_g = 3.77$ at 1550 nm for the isolated 0.5 μm and 3.0 μm waveguides, respectively. Assuming that the taper section contributes an intermediate group index that varies approximately linearly between the two waveguide widths, the expected effective group index of the composite differential path can be estimated from the length-weighted average

$$n_{g,\text{eff}} = \frac{\sum_i n_{g,i} L_i}{\sum_i L_i}, \quad (20)$$

which yields $n_{g,\text{eff}} \approx 3.97$. The experimentally extracted value obtained from the MZI8 spectrum is $n_g = 3.859$, corresponding to a deviation of only about 2.8% from the simple composite-waveguide prediction.

The measured value also lies between the compact-model predictions of the individual 0.5 μm and 3.0 μm waveguides, as expected. The remaining discrepancy is attributed primarily to the simplified treatment of the taper section and to the fact that the local modal properties evolve continuously along the adiabatic transition rather than remaining constant.

E. Directional Coupler Characterization

Finally, Figs. 14 and 15 show the Ref-normalized transmission spectra of the two output ports of the ADC and BDC,

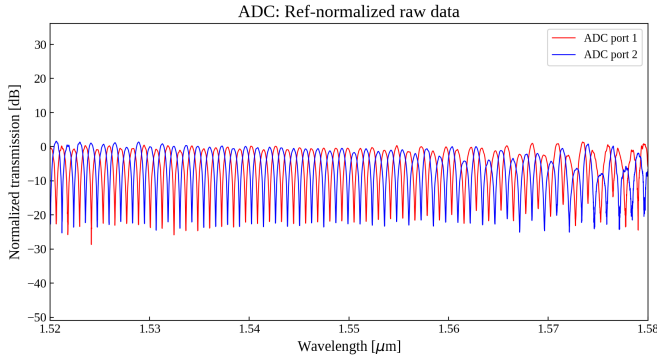


Fig. 14. Reference-normalized transmission spectra of the two output ports of the adiabatic directional coupler (ADC), a nominally 50:50 coupler. The fringes of port 1 (red) and port 2 (blue) are shifted by approximately half a period relative to each other, consistent with the expected 180° phase relationship between the two output ports of a 50:50 directional coupler.

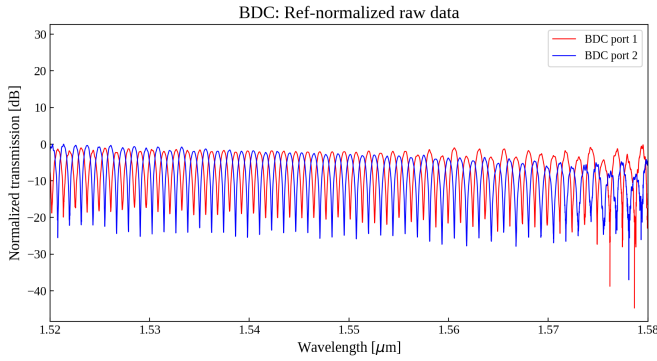


Fig. 15. Reference-normalized transmission spectra of the two output ports of the broadband directional coupler (BDC), a nominally 50:50 coupler. As with the ADC, port 1 (red) and port 2 (blue) show fringes offset by approximately half a period, consistent with the expected 180° phase relationship between the two output ports of a 50:50 directional coupler.

respectively. Both directional couplers are designed with a nominal 50:50 splitting ratio, and both show the expected complementary fringe pattern in which the two output ports are shifted by 180° relative to one another. No significant difference in overall spectral shape is observed between the ADC and the BDC.

VII. CONCLUSION

In this report, we presented the design, fabrication, and experimental characterization of silicon-photonic Mach-Zehnder interferometers on an SOI platform. Compact models of the effective refractive index were developed from eigenmode simulations of 500 nm single-mode and $3.0\ \mu\text{m}$ multimode silicon waveguides and were used to predict the group index, chromatic dispersion, and free spectral range (FSR) of MZI devices.

A dedicated test element group containing MZIs with path-length differences from 25 to $400\ \mu\text{m}$, a hybrid multimode MZI, and directional couplers was fabricated through the NanoSOI electron-beam lithography process and experimen-

tally evaluated. By fitting the measured transmission spectra with an analytical MZI model, the effective index, group index, and dispersion were successfully extracted. The experimentally obtained group indices and FSRs showed excellent agreement with the analytical predictions and fabrication-corner simulations, confirming both the validity of the compact-model approach and the robustness of the fabrication process. Nominally identical devices exhibited only a small effective-index variation of approximately 0.006, demonstrating good intra-chip reproducibility.

The hybrid MZI incorporating a widened $3.0\ \mu\text{m}$ waveguide section yielded a measured group index consistent with the length-weighted prediction based on its composite waveguide geometry. These results demonstrate that the proposed compact-model methodology provides an effective framework for the design, analysis, and characterization of silicon-photonic interferometric devices.

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